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Assessment of the aerobic preparation and bottom ash addition as pretreatment steps before landfilling: Impact on methanogenesis kinetics and leachate parameters

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ABSTRACT

This work focuses on assessing the impact of two types of waste pretreatment: addition of bottom ashes and aerobic pretreatment on both the onset and kinetics of methanogenesis and the evolution of different parameters in the leachate. It also studies the correlation between methane production and the different parameters measured in the leachate produced. A total of six 68-L pilots were thus used with fresh municipal solid waste (MSW) shredded to a 40-mm size. After 14 months of landfilling, the control has produced less than 10 NL kg^{-1} DM, which corresponds to around 7% of its biochemical methane potential (BMP). Nevertheless, on one hand for aerobically pretreated waste, the lag phase before the onset of methanogenesis is significantly reduced to 0.9 month compared to more than 1 year for the control. In addition to that, on average 110 NL kg^{-1} DM (90% of the BMP) is produced within around 6.5 months. On the other hand, the waste with added bottom ash shows a slight improvement of the lag phase over the control for one of the duplicate: 6.1 months of lag phase. At this stage, on average of 26 NL kg^{-1} DM waste are detected (22% of the BMP) no final conclusion concerning the impact of bottom ashes could be made. The data obtained for the leachate parameters agrees with the observations on methane production. Statistical correlation study shows that the two components of the corrected PCA interpret 76% of the variability of the data: SUVA (specific UV absorbance at 254 nm) and HPI* (% of hydrophilic compounds) are identified as interesting parameters for following up the biodegradation in landfill conditions.

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1. Introduction

According to the French legislation (Arrêté ministériel du 09/09/1997), monitoring and treatment of both leachate and biogas produced in the landfill are obligatory during site operation and extend for an aftercare period of at least 30 years. This legislation joins, at the European level, the council directive 99/31/EC which also aims at preventing or reducing as far as possible the negative impacts on the environment from landfilling through the introduction of stringent technical requirements such as waste treatment. And more recently the European Directive 2008/99/CE promotes the increase in the waste material recovery rate. Unfortunately despite all these challenges, municipal solid waste (MSW) landfilling remains one of the most common waste treatment processes in France. Published data of the French environmental agency (ADEME) show that, in 2006, 48 million tons of MSW were produced with almost 50% diverted to landfills. It is also the case in Italy where even with the high rate of selective collection, 40% of the waste goes to the non-differentiated MSW fraction with an organic content varying between 20% and 25% (Di Maria et al., 2010). In China, 80% of the generated MSW is landfilled (Yue et al., 2011). In this latter case, the organic fraction is predominant and leads most of the times to an acidification in the landfill and consequently longer lag phases before methanogenesis and longer periods for the waste (Shao et al., 2005). In summary worldwide important amounts of organic matter still go to the landfill with important methane and other green house gas emissions potentials.

In Europe and more specifically in Austria and Germany, the mechanical biological treatment (MBT) has been widely used in order to stabilize the waste before landfilling. Biostabilization processes of 2–6 months have been set up keeping in mind that according to De Gioannis et al. (2009) between 80% and 91% reduction of biogas potential can be achieved after 8 and 15 weeks of pretreatment respectively. However, in the European Directive 2001/77/CE, which promotes the increase in the contribution of renewable energy sources to electricity production, the biogas produced by the degradation of waste in the landfills is cited as a source of

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renewable energy. Within this new context of increasing environmental concern and promoting renewable energies, it becomes essential to invest in the optimization of current practices of conventional landfilling and thus modify the approach to the MBT. Waste pretreatment becomes again one key factor with high potential. An interesting consequence has thus been the shift towards a new MBT concept in Italy which reduces the biostabilization period to 2–4 weeks only (Scaglia et al., 2011). Even for the bioreactor landfill, also in Italy, waste pretreatment is looked at to reduce operational problems on site and increase the biogas production (Di Maria and Valentini, 2011). In China, waste pretreatment is also being approached as the solution to limit acidogenesis and thus shorten the lag phase in the landfill (Yue et al., 2011).

In this perspective this work, part of the project BIOPTIME (ANR Precodd 2006) funded by the French National Agency on Research, focuses on studying two types of pretreatments before landfilling: addition of bottom ashes and aerobic pretreatment.

On one hand, bottom ashes are the heterogeneous solid residues remaining after municipal solid waste (MSW) incineration. According to the French Environment and Energy Management Agency (ADEME), 21% were co-disposed with MSW in landfills in 2002, which represent more than 500,000 tons of these residues. In addition to the fact that important amounts of bottom ashes are available, mixing those particles with the waste is expected to generate buffering capacity in the medium due to the alkalinity of the bottom ashes. This latter can positively impact the enzymatic activity of the microorganisms (Lo et al., 2009) thus accelerating the process of biodegradation (Wang et al., 2009). In addition to that, the hydrolysis of polymers contained in the waste results in the production of volatile fatty acids that may cause a drop in the pH (Siegert and Banks, 2005; Wang et al., 2009). Generally, the consumption of VFA by methanogenic microorganisms counterbalances this accumulation but if the rate of hydrolysis is too strong, the pH may drop, inhibiting the overall degradation process. The alkalinity brought by the bottom ashes may therefore counterbalance the acidity generated during the initial fermentation stage and thus accelerate the overall degradation process (Johnson and Furrer, 2002; Guimaraes et al., 2006; Polettini et al., 2007).

On the other hand, Johansen and Bakke (2006) mentioned that even the injection of small amounts of air allows an increase in the hydrolysis rate of waste due to the aerobic activity of the microorganisms. Therefore, the major target of the aerobic pretreatment is the initiation of the hydrolysis that could be a limiting kinetic step for this type of waste. Literature data shows that lignocellulosic compounds, highly abundant in waste, are not easily degradable by microorganisms (Jash and Ghosh, 1996; Ress et al., 1998). The aerobic pretreatment aims thus at initiating the degradation of those types of compounds. Enzymes produced by aerobic fungi can attack the cellulose fibers and allow a better colonization and by that a better degradation of the waste when this latter is placed under anaerobic conditions.

Taking in account these advantages, this work studies the effect of each pretreatment on the onset and kinetics of methanogenesis and the leachate quality. It focuses also on the correlation between methane production and the different parameters measured in the leachate produced in order to find adequate follow-up parameters of the methanogenesis.

2. Materials and methods

2.1. Experimental setup

The pilots used in this study are cylindrical units made up of opaque polyvinyl chloride (PVC) (Fig. 1). They are all air tight with a total working volume of 68 L (H = 63 cm and D = 37 cm). The leachate collection system for the pilots is made up of an inclined bottom and a stainless steel perforated plate which allows the downward flow of the leachate from the waste (Fig. 1). At the lowest point of the leachate reservoir, a valve allows the easy collection of the leachate.

The cover of each pilot is equipped with five valves: a pressure control valve, a valve connecting the inner atmosphere to a micro-gas chromatograph (Varian CP-4900), a valve connected to a drum-type gas meter (Ritter model TG1), a 3-way valve that allows injecting liquid into the system without any gas entry and finally a security valve.

During the experiments, the pilots are placed in a room at 35 °C. The inner atmosphere of each pilot is constantly kept in slight overpressure. The pressure control valve of every pilot is connected to an automatic system that whenever the pressure builds up in the system and reaches 0.3 bar, the pressure opens till the pressure drops to 0.1 bar.

In this study the used feedstock is a fresh MSW obtained from an industrial municipal solid waste treatment plant, where it is crushed with an industrial mill and then screened to obtain the fraction under 40 mm. After this pretreatment, the waste is sent on the same day from the industrial plant and placed directly in the 68-L pilots or sent for aerobic pretreatment. On average, 14.2 ± 0.8 kg of dry matter (DM) were placed in each pilot with almost 50% moisture content. The initial VSs (volatile solids) content was measured at 52 ± 5% of TS (total solids). Its average biochemical methane potential (BMP) was measured at 140 NL CH_4/kg dry matter (DM) of waste. For all samples, the BMP was determined on dried and grinded samples. All tests were performed in 0.5 L serums bottles, in mesophilic conditions and with a substrate/inoculum ratio ranging from 0.2 to 0.5 g of $VS_{substrate}$ per gram of $VS_{inoculum}$. The cumulated biogas production was measured during 44 days and analyzed by micro-gas chromatography. The tests were carried out in duplicate. More details about the method are available in Angelidaki et al. (2009) and Hansen et al. (2004).

In addition to that, Table 1 presents the general composition of the MSW according to the guidelines of the national household characterization survey conducted in 1993 (ADEME, 1995).

Previous experiments (data not published) have shown that when the waste is placed in the pilots even at 50% humidity, the onset of methanogenesis did not occur for a control even after 1 year of landfilling. It was therefore decided to saturate the waste at its placement in the pilot with a leachate collected from a landfill operated by Veolia Environnement in the Ile de France region (78). Over 30 L of fresh leachate were injected in each pilot. They were kept in contact with the waste for 2 h and then drained by gravity; almost 70% of the injected leachate was recovered by

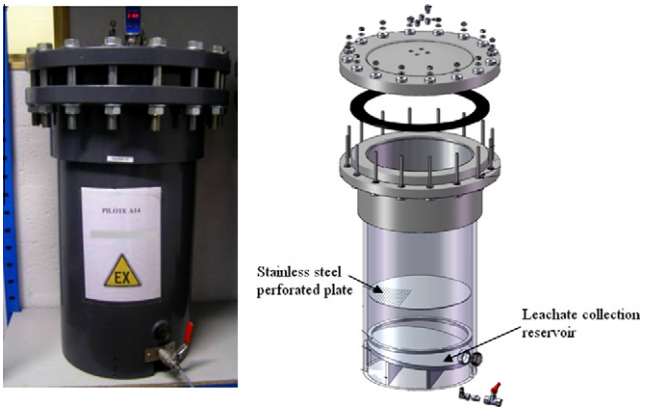


Fig. 1. Photo of a 68 L pilot used in the study (left) and schematic presentation (right).

Table 1
General composition of the fresh MSW used in the pilots (wet weight basis).

Fraction	%
Easily biodegradable and fine particles	58.1
Paper and cardboard	13.3
Plastic	9.5
Glass	5.4
Metals	0.4
Textile	2.1
Others	11.2

gravity. Later the headspace was purged with nitrogen in order to make sure of complete anaerobic conditions. It is important to note that the remaining leachate was recovered from the pilots over an average period of 2 months following saturation/desaturation. Afterwards, almost no leachate was recovered from the pilots. Therefore 500 mL of distilled water are injected in each pilot on a monthly basis for the waste not to dry out since on-site the waste that has already been placed in an active cell still receives water from rainfall and humidity from newly added waste. In addition to that with the added water regularly, it is sure to have a minimum amount of leachate for analysis.

2.2. Pretreatments

As mentioned earlier, two types of pretreatment are tested in this work. On one hand, for the bottom ash addition, the optimal amount has been previously optimized within the context of the project (Boulanger et al., 2009a); the rate of addition was found to be less than 15% of bottom ashes per kg of dry waste. In this case, bottom ashes were mixed with the MSW in an automatic rotating drum reactor until homogenization of the mixture and then the mixture was placed in the 68-L reactors. Note that no VS content is brought in by the bottom ashes; these incineration particles are mainly made up of mineral content.

On the other hand, the aerobic pretreatment consisted of a composting phase which corresponds to a stage during which the degradation occurs aerobically (air is forced into the waste matrix). This step is accompanied by a complete monitoring of the gas emissions mainly those of carbons such as total hydrocarbons (THCs) and carbon dioxide as well as the nitrogenous compounds. The composting process was kept for less than 5 days in order to achieve a total carbon loss level between 10% and 20% carbon loss per kg of dry waste. The level of aerobic pretreatment has already been optimized on a reconstituted MSW in the 68-L pilots in Mansour et al. (2010).

In total, six pilots are followed: a control "C", two duplicate pilots with an aerobic pretreatment "A_x" and "A_y", two duplicate pilots with added bottom ashes "B_x" and "B_y" and one pilot with both an aerobic pretreatment and addition of bottom ashes "AB". In fact, the pilot AB is used to see if there is any synergistic effect of both pretreatments.

2.3. Gas analysis and modeling

The volume of the biogas produced is measured using a drum-type Ritter gasmeter (type TG1) and the quality of the biogas is determined using a Varian micro-gas chromatograph (CP 4900) with helium as a carrier gas. Percentages of CH_4, H_2S, CO_2, O_2, N_2 and H_2 are measured on a daily basis.

In order to characterize and compare the control and the different pretreatment pilots, cumulative production curves are modeled using a modified Gompertz equation (Zwietering et al., 1990; Lay et al., 1998a,b; Zhu et al., 2009). The modified equation is described below:

$$\mathrm{Pcm}(t) = \mathrm{Pcm}(\infty) \times \left(\exp\left(-\exp\left(\frac{\mathrm{Vm} \times e}{\mathrm{Pcm}(\infty)} \times (\lambda - t)\right)\right) + 1\right)$$

where $\mathrm{Pcm}(\infty)$ corresponds to cumulative methane production at the end of the experiment (NL kg^{-1} DM); Vm, maximum rate of methane production (NL kg^{-1} DM day^{-1}); e, exponential of 1; λ, lag phase (day); t, time (day).

2.4. Leachate analysis

An extensive follow-up of the leachate is conducted on a monthly basis on all pilots for nine different parameters. Both pH (CyberScan pH 510, Eutech instrument) and conductivity (Tetracom 325, LF 538) are followed with time. Dissolved organic carbon (DOC) is measured using a TEKMAR DOHRMANN (model Phoenix 8000) carbon analyzer and chemical oxygen demand (COD) using the closed reflux colorimetric method (NFT 90-101). UV 254 nm is measured using a doubled-beam spectrophotometer (PharmaSpec 1700 from Shimadzu) with 1 cm-long quartz cells. Total alkalinity (TA) and volatile fatty acids (VFAs) are evaluated through a potentiometric dosage. Ammonia (NH_4^+) is measured using a colorimetric method with a multi-parameter analyzer AQ2 from Seal analytical and total nitrogen (Nt) according to the French norm NF EN 25663 T90-110.

In addition to the parameters mentioned above, the characterization of the organic matter is carried out by fractionation according to the hydrophobic character of the organic molecules (Weishaar et al., 2003). Humic-like compounds are eliminated by filtration through 0.45 µm membrane. Then, humic-like acids (HA^*) are separated by acidification at pH 2 and filtration through a 0.45 µm membrane. The samples are passed through DAX-8 and XAD-4 resins in series to separate according to the hydrophobic character the following fractions: hydrophobic-like substances (HPO^*), transphilic-like substances (TPH^*) and hydrophilic-like substances (HPI^*).

Based on the data above three indices are calculated. The SUVA index (the ratio of the UV absorbance to the DOC) is calculated to estimate the aromatic character of the organic molecules which reflects the state of biodegradation of the waste matter. The HS^* which is the sum of the HA^* and the HPO^* corresponds to the part of hydrophobic compounds. They can be considered as weakly biodegradable organic compounds. Finally the ratio HA^*/FA^*, in which FA^* stands for fulvic like-acid and is assimilated to the HPO^* fractions, is calculated. HA^*/FA^* is used to evaluate the state of stabilization of the waste: the higher the ratio, the higher the state of stabilization.

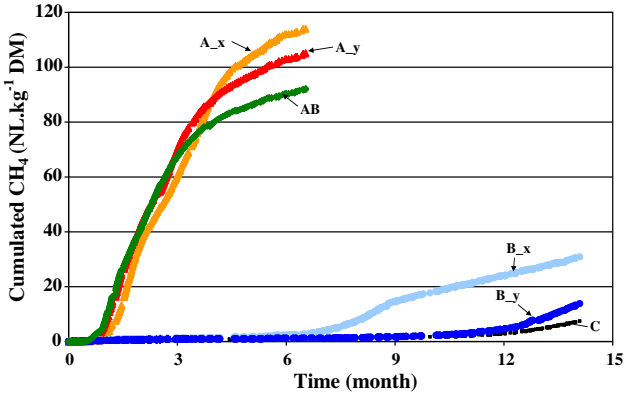


Fig. 2. Evolution of the methane production (NL/kg DM) in the different pilots with time: control C, aerobic pretreatment A_x and A_y, addition bottom ashes B_x and B_y and aerobic pretreatment and with addition of bottom ashes AB.

Table 2
Gompertz parameters for the pilots: C, A_x, A_y, B_x, B_y and AB (italic values are not final since the gas production is not stable).

Pilot	λ (months)	Vm (NL kg^{-1} DM day^{-1})	Pcm(∞) (NL kg^{-1} DM)
C	12.6	*0.13*	*70*
A_x	1.1	1.09	119
A_y	0.8	1.15	104
B_x	6.1	*0.14*	*62*
B_y	11.4	*0.16*	*64*
AB	0.8	1.15	89

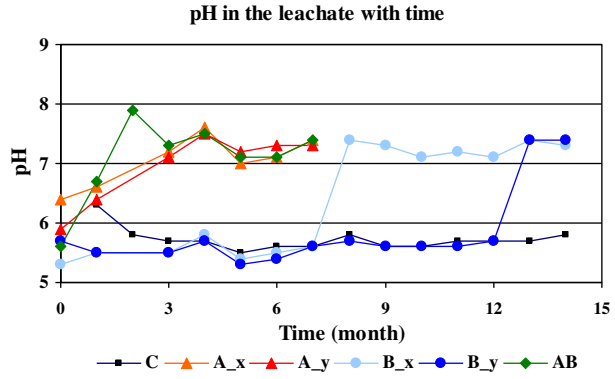


Fig. 3. Evolution of the pH of the leachate with time in all pilots.

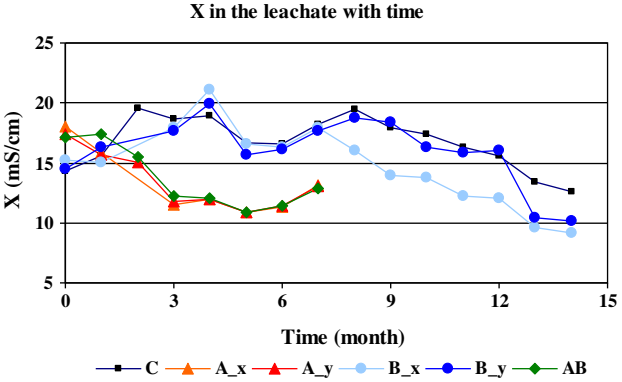


Fig. 4. Evolution of the conductivity EC of the leachate with time in all pilots.

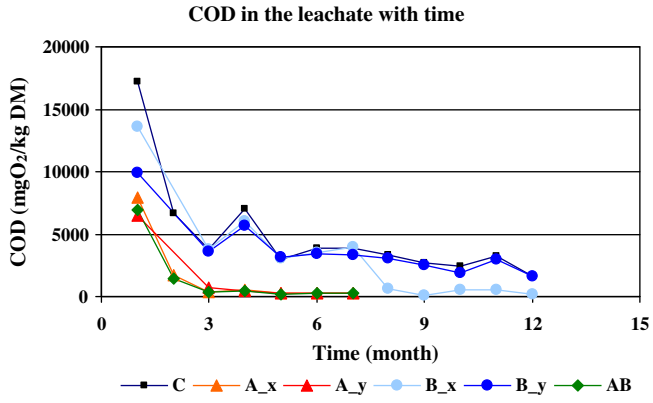


Fig. 5. Evolution of the chemical oxygen demand (mg O_2/kg DM) with time in all pilots.

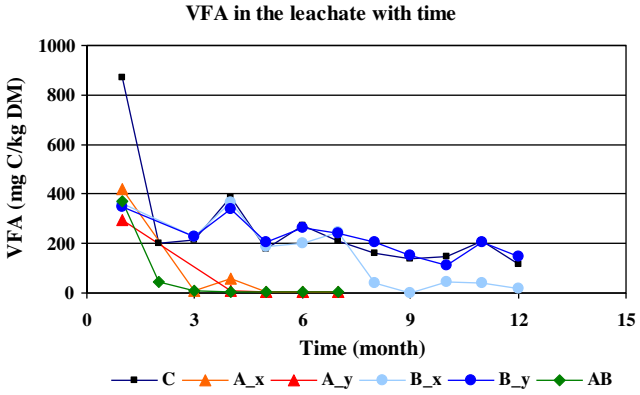


Fig. 6. Evolution of the concentration of volatile fatty acids (VFAs) (mg C/kg DM) in the leachate with time for all the pilots.

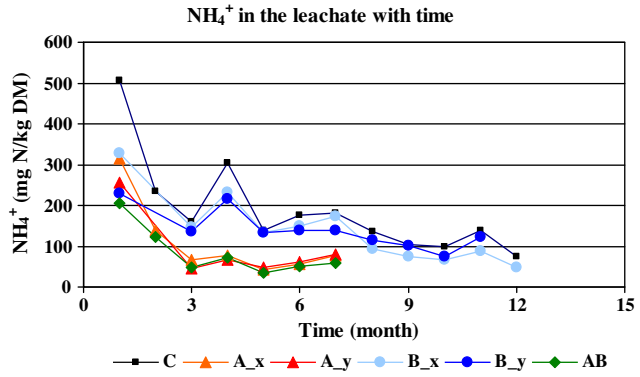


Fig. 7. Evolution of the concentration of NH_4^+ (mg N/kg DM) in the leachate with time for all the pilots.

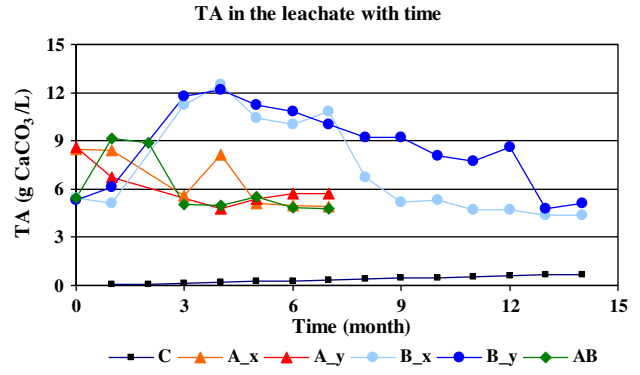


Fig. 8. Evolution of the total alkalinity TA in the leachate with time for all pilots.

2.5. Statistical approach

All statistical analyses are preceded by the determining if the data follow a normal distribution using the Shaphiro–Wilk test. Since it was not the case, all data were normalized using the Box–Cox method (Box and Cox, 1964). Given the nature of the data, it was most suitable to conduct a principal component analysis (PCA) for the interpretation. It is so because the PCA is an exploratory statistical method based a factorial analysis (Jackson, 2003). It is based on the visualization of the information given in a data set in the form of scattered points in geometrical spaces. The principal components are the orthogonal axes around which the

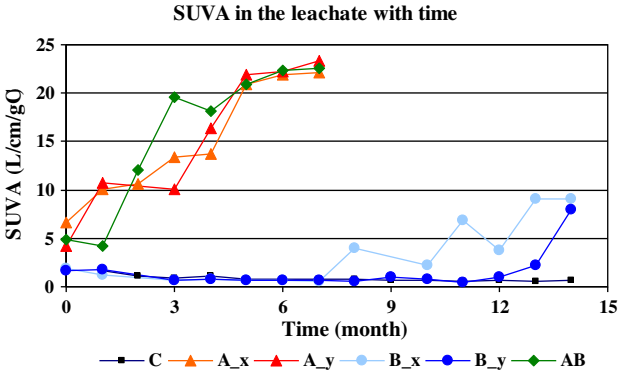


Fig. 9. Evolution of the SUVA index in the leachate with time.

data set is structured taking into account the maximum variability. All statistical analyses were performed using Statistica 9.1.

3. Results and discussion

3.1. Impact on methanogenesis kinetics

Methane production presented below (Fig. 2) is given in terms of NL of methane per kg of DM of waste since added bottom ashes consist mainly of mineral matter that does not contribute to the methane potential of the waste; this presentation allows a comparison of the different pilots on an equal basis.

The figure above (Fig. 2) shows that the pilots are divided into two distinct groups: the first one corresponds to the three pilots with the aerobic pretreatment and the second one the control and the two pilots with added bottom ashes. Results thus show a significant advantage of the aerobic pretreatment over the control and bottom ashes addition. In 6.5 months of landfilling, methane production is stabilized for the three pilots A_x, A_y and AB; respectively 98%, 90% and 79% of the initial methane potential are expressed. Nevertheless these values should be corrected since the step of saturation/desaturation at the waste placement has

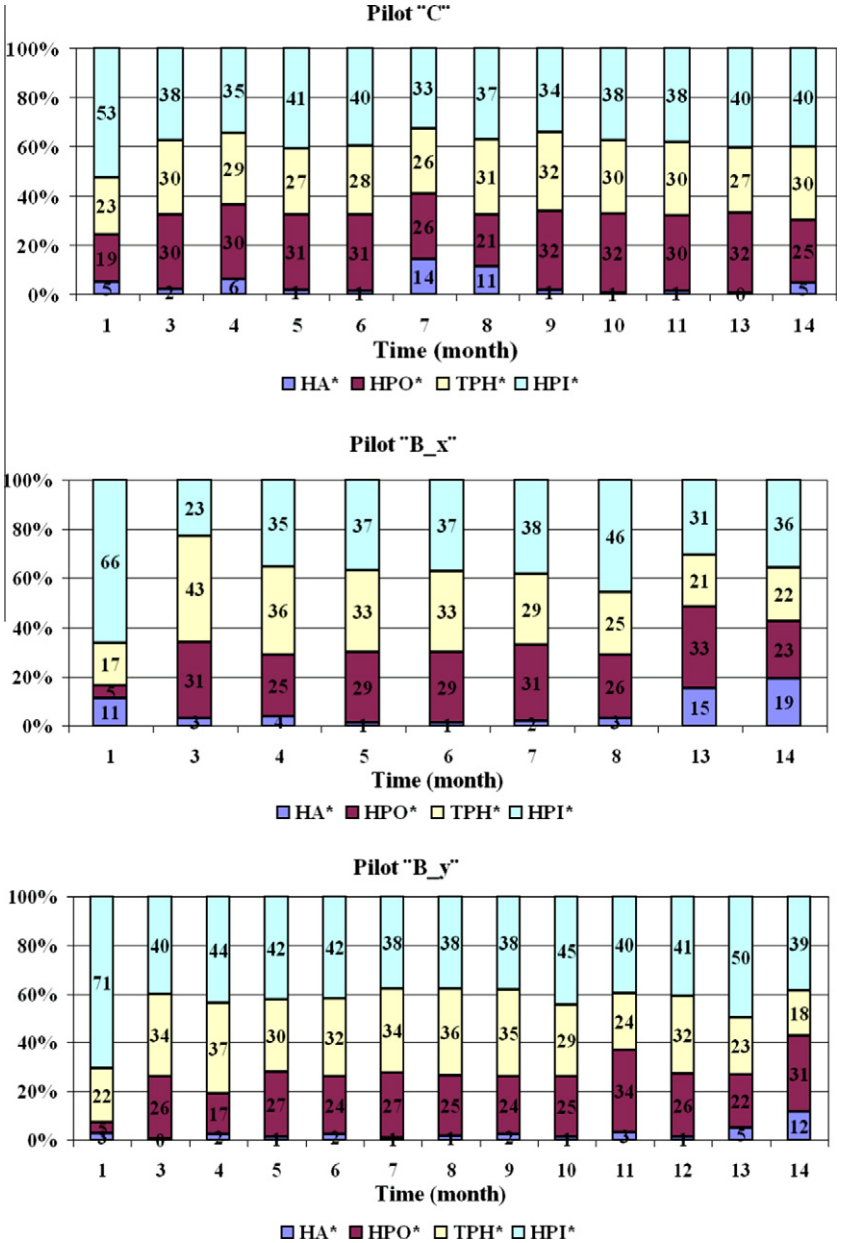


Fig. 10. Evolution of the fractions HA*, HPO*, TPH* and HPI* in the pilots C, B_x and B_y.

removed some inherent COD of the waste matrix. After conversion of the COD into methane equivalent, 100%, 100% and 91% of the initial methane potential (corrected values) are expressed for the above mentioned pilots. Moreover, it seems that adding bottom ashes to aerobically treated waste did not have any significant impact on the methane production. The differences among the three pilots could be simply due to experimental errors and most importantly to the heterogeneity of the waste matrix.

Contrary to the aerobic pretreatment, bottom ash addition did not show interesting results under unsaturated conditions. After 14 months of landfilling, only 28% and 12% of the methane potential was produced for B_x and B_y respectively (43% and 28% corrected values). Interestingly the pilot B_x shows an onset of the methanogenesis earlier than that of its duplicate B_y and the control; an observation that could be explained by the heterogeneity of the matrix. Unfortunately, the results obtained by Boulanger et al. (2009a,b), under saturated conditions, in which bottom ash addition shows very interesting results in terms of kinetics could not be confirmed with the 68 L-pilots. The action mode of the bottom ashes might thus thrive more under saturated conditions.

Since the methane production of the pilots C, B_x and B_y is in its early stages, both Vm and Pcm values for these pilots should be thus taken carefully (Table 2).

The results of the lag phase λ show clearly that aerobic pretreatment presents a significant advantage over the control and bottom ash addition. The onset of methanogenesis is on average 0.9 month after waste placement for the A_x, A_y and AB pilots. It is thus almost 11.2 times longer on average for the remaining pilots C, B_x and B_y taking into account the significant difference in λ between the duplicate pilots B_x and B_y (6.1 months vs. 11.4 months).

These observations could be explained by the fact that at least part of the hydrolysis step takes place during the aerobic pretreatment. Once the waste is placed under anaerobic conditions the saturation/desaturation with the leachate can remove an important fraction of the remaining volatile fatty acids of the medium and thus methanogenesis could start almost immediately. However, for the

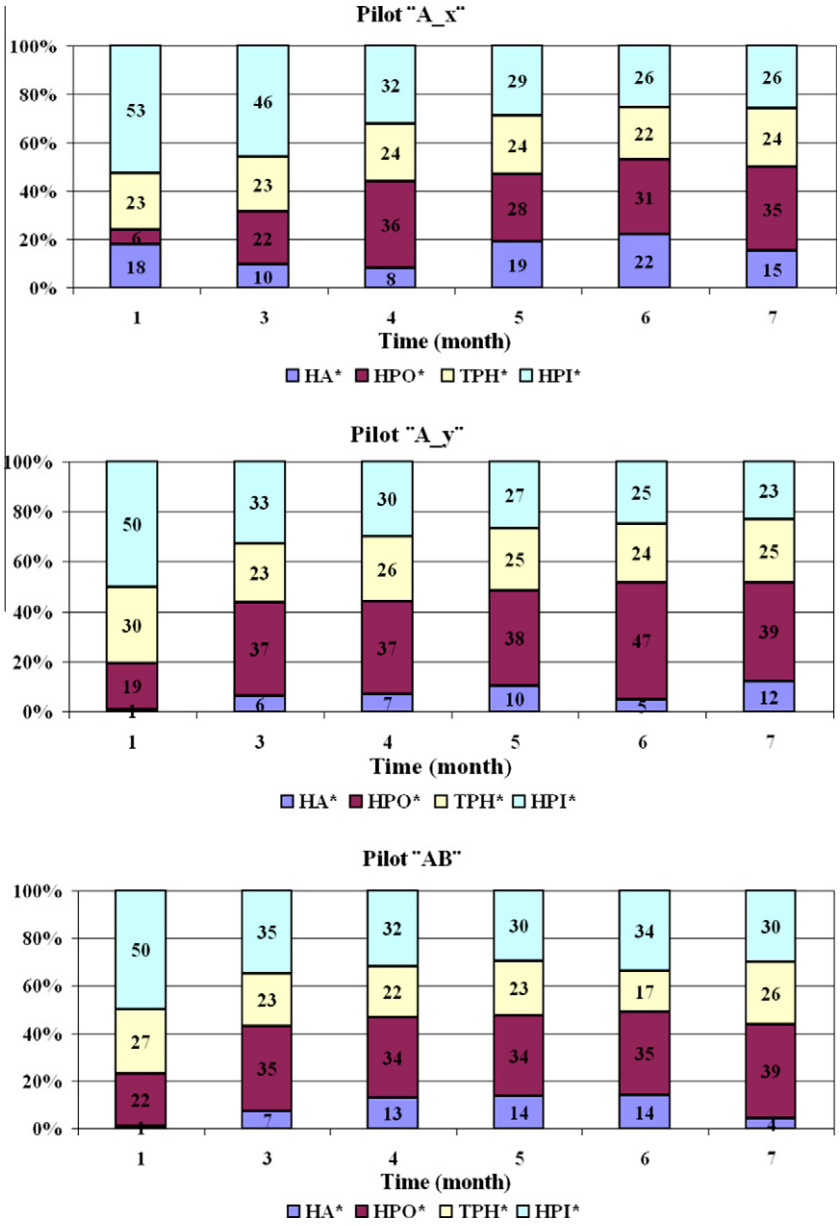


Fig. 11. Evolution of the fractions HA*, HPO*, TPH* and HPI* in the pilots A_x, A_y and AB.

remaining pilots C, B_x and B_y, hydrolysis occurs in the pilots and the waste does not have sufficient inherent buffering capacity to overcome the acidity: the pH of C and B_y is still below 6 while that of B_x remained below 6 till the 8th month of the experiment.

For Vm and Pcm, a comparison between the pretreatments could not be made. Nevertheless within the BIOPTIME project, Boulanger et al. (2009b) has run tests using the same levels of aerobic pretreatment but with reconstituted MSW and under saturated anaerobic conditions in 1 L-reactors. The authors found that the Vm values recorded at the 68-L scale are comparable to those presented by under saturated conditions: 1.15 for unsaturated vs. 1.19 for saturated conditions on average. Surprisingly the Pcm values are higher for the current work: 104 for unsaturated vs. 70 for saturated conditions; this could be explained by the fact that the waste used in the 68-L pilots is a real waste with inherent microflora that along with the saturation/desaturation with leachate step allows a better degradation profile.

3.2. Impact on leachate parameters

From the different parameters only the pH, EC, COD, TA, VFA, NH_4^+ and SUVA as well as the fractionation will be presented in details since DOC and COD follow similar profiles, the nitrogen released is mainly in the ammonia form and since SUVA is the ratio of DOC and UV.

As shown in Fig. 3, the evolution of the pH agrees with the evolution of the methane production presented in Fig. 2. The pH remains below 6 for the control up until 14 months of landfilling. For the pilots A_x, A_y and AB, the pH increases to neutrality between 2 and 3 months after landfilling. The increase to a pH of 7 for B_x appears at month 7 and that for B_y at month 12; these results show the variability among the duplicates with bottom ash addition and are in agreement with the variability observed on methane production for these pilots.

Fig. 4 shows that the decrease of conductivity EC is immediate for the three pilots with aerobic pretreatment. However, for the three remaining pilots, one inflexion point appears at month 7 for B_x and another at month 12 for B_y. The data thus agree with the methane produced and the pH values.

The COD data (Fig. 5) show that for the pilots A_x, A_y and AB the values of COD become almost negligible after 3 months of landfilling; the COD drops on average from 7100 mg O_2/kg DM to 480 mg O_2/kg DM, that is almost a 15-folds decrease in the COD value. For B_x the inflexion point is apparent at month 7, although the profile of B_y follows very closely that of the control. This observation could mean that although a methane production has started in the pilot B_y, methanogenesis remains in its early stages.

This latter observation is also identical when looking at the evolution of VFA with time for the different pilots (Fig. 6).

For the previous parameters, at least B_x was different from the control. While when looking into the ammonia concentration (Fig. 7), one can see only two distinct groups with C, B_x and B_y on one hand and A_x, A_y and AB on the other hand. In fact, Wens et al. (2001) and Yuen et al. (1995) indicate that an accumulation of NH_4^+ can lead to the inhibition of the degradation of the waste. This inhibition was observed by the authors for concentrations higher than 1500 mg/L. During the experiments the total quantity of NH_4^+ remains all the time below this value.

For all data shown so far, no significant difference was shown among the pilots B and C. However, when looking into the TA data (Fig. 8) an important difference appears for the pilots with bottom ash addition. The TA for the control remains below 0.7 g $CaCO_3$/L, however, for the pilots B_x and B_y it shows the highest values: the bottom ash addition did not have an impact on the pH of the leachate, however, an important impact on TA could be noticed. Surprisingly, this observation does not hold true for the pilot AB; one can assume that the impact of bottom ashes is masked by that of the aerobic pretreatment.

Finally, Fig. 9 shows the evolution of the SUVA index with time. The results shown agree with previously published data by François et al. (2006) and Feuillade et al. (2009). This index is characteristic of hydrophobic organic molecules as it provides indications on their aromaticity. Feuillade et al. (2009) considered that this index increases during waste degradation with values going from below 10 to above 30 L cm^{-1} g C^{-1}. As shown in Fig. 9, the SUVA index values for aerobically pretreated waste goes from around 5 to about 23 L cm^{-1} g C^{-1} while those of the remaining pilots are below 5 L cm^{-1} g C^{-1}. It is nevertheless noteworthy to point the inflexion point for B_x that correlates with the onset of the methanogenesis in this pilot.

In the following figures, the organic matter characterization in the leachate is presented with time (Figs. 10 and 11). This

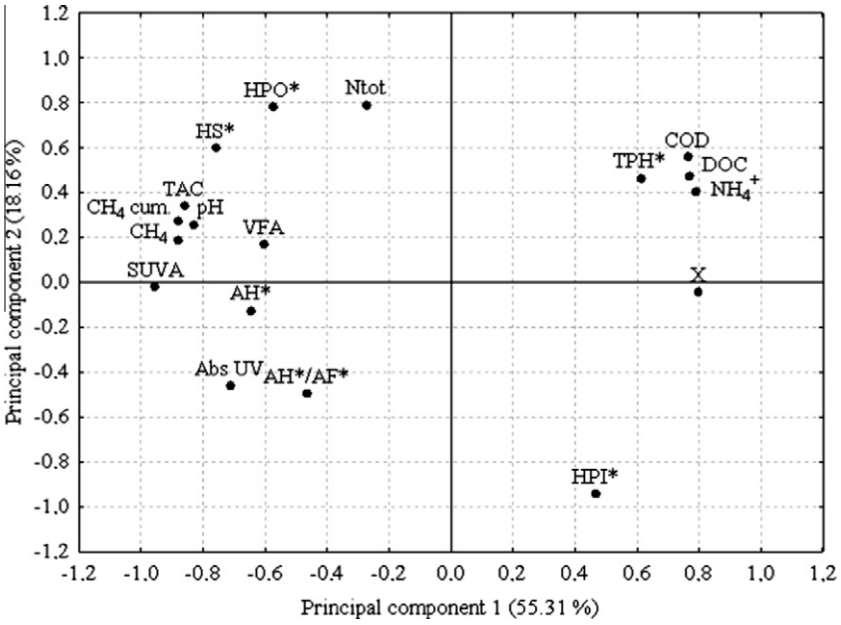


Fig. 12. Principal component analysis (PCA) for the different parameters in leachate analysis and methane production.

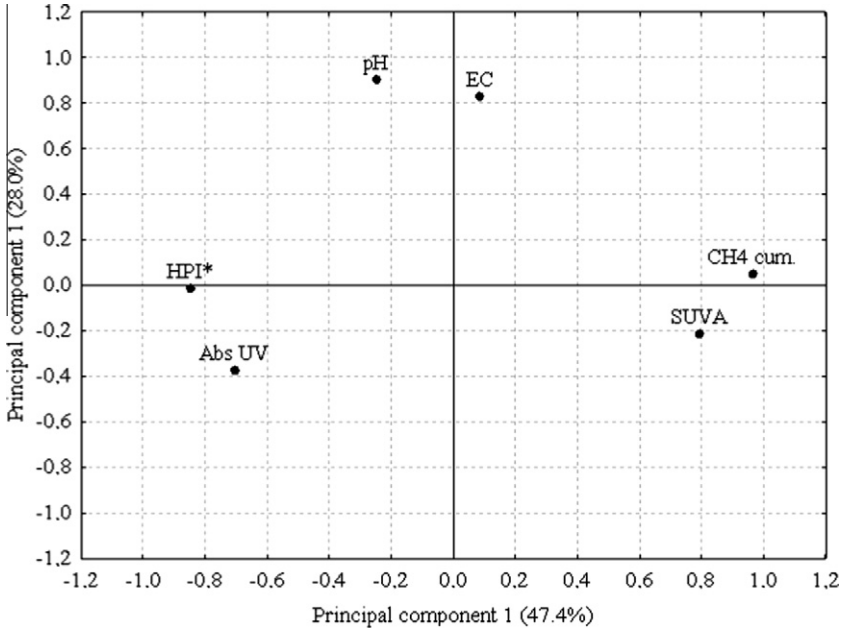


Fig. 13. Principal component analysis (PCA) for the different parameters in leachate analysis and methane production.

follow-up sheds the light on the evolution of the biodegradation state of the waste: the more hydrophobic the organic molecules are, the more advanced the state of degradation is (Berthe et al., 2008). In fact during the methanogenic phase, HPO* accumulate in the leachate (Blakey et al., 1992), later they decrease to the benefit of HA* increasing by that the ratio HA*/FA*.

In Fig. 10, the data show that the fractions of HA*, HPO*, TPH* and HPI* are stable with time mainly for pilots C and B_y. The data presented for the first month differs because it reflects mainly whatever remains of the landfill leachate that was introduced in the pilots at day 1. A variation appears starting month 9 for the pilot B_x; this agrees with the onset of methanogenesis in this pilot.

Contrary to Fig. 10, the data in Fig. 11 varies significantly as it evolves with time. For the pilots A_x, A_y and AB, the fraction of HA* is more important than the remaining pilots. The fractions of HPI* decrease with time while those of HPO* increase showing a shift towards more hydrophobic and thus more stable organic matter.

3.3. Statistical approach for the correlation between methane production and leachate parameters

Leachate follow-up of the 68 L pilots has been extensive therefore an important objective of this work is to identify leachate parameters that could be used as indicators of the biodegradation through a statistical approach.

In this perspective, primary analysis showed that the data did not follow a normal distribution therefore all parameters were transformed using the Box–Cox method. Since the data set consisted of a large number of interrelated variables, the PCA was achieved in order to reduce the number of the observed variables while retaining as much as possible of the variation present in the data set (Jolliffe, 2002). Therefore, a first PCA (Fig. 12) was generated using all the data set. The two principal components of this PCA were able to interpret in total almost 74% of the total variability of the data. But the number of parameters was only slightly reduced.

It was therefore decided to conduct another PCA while taking into account only the data set that corresponds to the stage after the initiation of the methanogenesis. One major reason is that for the pilots C, B_x and B_y there was no methane production for an important period of time, only null values are generated for this parameter and this can easily distort the analysis.

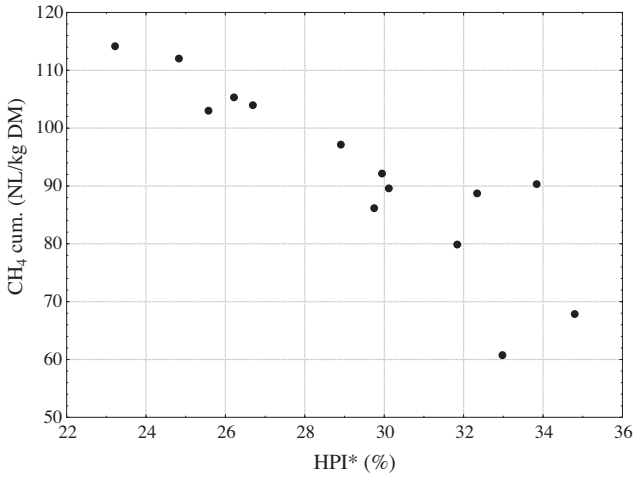


Fig. 14. Significative correlation between cumulative CH_4 production and % HPI* in the leachate.

The second PCA (Fig. 13) reduced thus more significantly the number of variables: SUVA, HS*, HPO* are directly correlated to the cumulative CH_4 production while EC, pH, UV absorbance and HPI* are negatively correlated. The two components explain still 73% of the variability of the data.

From this analysis, the first indicator to be confirmed as a relevant indicator of the biodegradation is SUVA index. However, in order to confirm the correlation between CH_4 cum and HPI*, the HS* variable was removed as well as two data points from the scores that distorted the distribution. HPO* was also removed since it did not show an interesting impact on neither components of the PCA. The two components of the new PCA explain 76% of the variability and express better the correlation between the two parameters. This covariance is confirmed by a Coefficient of Pearson of -0.88 (Fig. 14). HPI* is confirmed as an interesting parameter of follow-up.

4. Conclusions

Aerobic preparation and bottom ash addition have been tested as pretreatment steps of a French MSW before landfilling in 68 L-pilots. After 14 months of landfilling, the control shows no significant production of methane. Nevertheless, on one hand for aerobically pretreated waste, the lag phase before the onset of methanogenesis is significantly reduced to less than 1 month compared to 1 year for the control. In addition to that, more than 90%, on average, of the BMP of the landfilled waste is expressed in around 6.5 months. This is accompanied by an important rate of methane production that goes up to 1.17 NL CH_4 kg^{-1} DM day^{-1}. This aerobic pretreatment has thus proven its efficiency and the scaling-up becomes the future industrial challenge to look into. On the other hand, the waste with added bottom ash shows a slight improvement of the lag phase over the control for one of the duplicates (6.1 months vs. 12.6 months for the control) and of the biogas produced (26% for the bottom ashes and 7% for the control). At this level of the experiments no final conclusion could be made for the effect of bottom ashes until stabilization of the biogas production.

The results of the leachate follow-up for COD, pH, EC, TA, VFA and ammonia agree with the results of the methane production. Using all data sets, the two components of the corrected PCA interpret 76% of the variability of the data. Both the SUVA and the HPI^* appear as interesting parameters for following up the biodegradation in landfill conditions.

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